

University of Colombo School of Computing (UCSC)
University of Colombo
SCS 1307 – Probability and Statistics
Assignment 01 – Semester I – 2026
Answer Script

Index Number: _____

Question 1

(i)

A password is formed by selecting one digit from 0–9 and one letter from A–Z.

Even digits: $\{0, 2, 4, 6, 8\}$, so there are 5 even digits out of 10.

$$P(\text{even digit}) = \frac{5}{10} = \frac{1}{2}$$

Vowels: $\{A, E, I, O, U\}$, so there are 5 vowels out of 26 letters.

$$P(\text{vowel}) = \frac{5}{26}$$

Since the digit and letter are selected independently:

$$P(\text{even digit and vowel}) = \frac{1}{2} \times \frac{5}{26} = \boxed{\frac{5}{52} \approx 0.0962}$$

(ii)

Two random integers from 1 to 6 (with replacement) are generated. Total outcomes = $6 \times 6 = 36$.

Outcomes with sum ≥ 9 :

- Sum = 9: $(3, 6), (4, 5), (5, 4), (6, 3)$ – 4 outcomes
- Sum = 10: $(4, 6), (5, 5), (6, 4)$ – 3 outcomes
- Sum = 11: $(5, 6), (6, 5)$ – 2 outcomes
- Sum = 12: $(6, 6)$ – 1 outcome

Total favourable outcomes = $4 + 3 + 2 + 1 = 10$.

$$P(\text{sum} \geq 9) = \frac{10}{36} = \boxed{\frac{5}{18} \approx 0.2778}$$

(iii)

Let P = event student uses Python, R = event student uses R.

Given: $P(P) = 0.70$, $P(R) = 0.45$, $P(P \cap R) = 0.30$.

$$P(R | P) = \frac{P(P \cap R)}{P(P)} = \frac{0.30}{0.70} = \boxed{\frac{3}{7} \approx 0.4286}$$

Question 2

Given: $P(A) = 0.7$, $P(B) = 0.5$, $P(B | A) = 0.6$.

(i) $P(A \cap B)$

$$P(A \cap B) = P(B | A) \cdot P(A) = 0.6 \times 0.7 = \boxed{0.42}$$

(ii) $P(A \cup B)$

$$P(A \cup B) = P(A) + P(B) - P(A \cap B) = 0.7 + 0.5 - 0.42 = \boxed{0.78}$$

(iii) $P(A | B)$

$$P(A | B) = \frac{P(A \cap B)}{P(B)} = \frac{0.42}{0.5} = \boxed{0.84}$$

(iv) Independence of A and B

A and B are independent if and only if $P(A \cap B) = P(A) \cdot P(B)$.

$$P(A) \cdot P(B) = 0.7 \times 0.5 = 0.35$$

Since $P(A \cap B) = 0.42 \neq 0.35$, the events A and B are **not independent**.

Question 3

Let A , B , C denote the events that a request is processed by Server A, B, C respectively, and F denote the event that a request fails.

$$P(A) = 0.40, \quad P(B) = 0.45, \quad P(C) = 0.15$$

$$P(F | A) = 0.03, \quad P(F | B) = 0.05, \quad P(F | C) = 0.08$$

(i) $P(B \cap F^c)$

$$P(B \text{ and not fail}) = P(B)(1 - P(F | B)) = 0.45 \times 0.95 = \boxed{0.4275}$$

(ii) $P(F)$ – Law of Total Probability

$$P(F) = P(A)P(F | A) + P(B)P(F | B) + P(C)P(F | C)$$

$$P(F) = (0.40)(0.03) + (0.45)(0.05) + (0.15)(0.08)$$

$$P(F) = 0.0120 + 0.0225 + 0.0120 = \boxed{0.0465}$$

(iii) $P(C | F)$ – Bayes' Theorem

$$P(C | F) = \frac{P(C)P(F | C)}{P(F)} = \frac{(0.15)(0.08)}{0.0465} = \frac{0.0120}{0.0465}$$

$$P(C | F) \approx \boxed{0.258}$$

Question 4

Requests are independent, each with probability of being invalid $p = 0.15$, and $n = 10$ requests are received. Let X be the number of invalid requests.

(i) Distribution of X

$$X \sim \text{Binomial}(n = 10, p = 0.15)$$

$$P(X = x) = \binom{10}{x} (0.15)^x (0.85)^{10-x}, \quad x = 0, 1, \dots, 10$$

(ii) $P(X = 2)$

$$P(X = 2) = \binom{10}{2} (0.15)^2 (0.85)^8$$

$$\binom{10}{2} = 45, \quad (0.15)^2 = 0.0225, \quad (0.85)^8 \approx 0.27249$$

$$P(X = 2) = 45 \times 0.0225 \times 0.27249 \approx \boxed{0.2759}$$

(iii) Mean and Variance of X

$$\text{Mean: } E[X] = np = 10 \times 0.15 = \boxed{1.5}$$

$$\text{Variance: } \text{Var}(X) = np(1 - p) = 10 \times 0.15 \times 0.85 = \boxed{1.275}$$

_____ End _____